DC Circuits — Reference

Textbook: Ulaby, Maharbiz, Furse, *Circuit Analysis and Design*, 3rd ed., chapters 1–3. Gray tags like [book 2-1] point to the book section. The book writes voltage as υ (upsilon). These notes write v.

how to read this note

Every method is a chain of steps. Each step is a template with empty colored slots.

Do	How
1. find the pieces	Each topic opens with a PIECES table. It says where each piece is in the circuit or the problem, and its color.
2. fill the slots	A piece goes in the slot of the same color: own R goes in own. In a diagram, a colored element is that piece.
3. one box at a time	Each box is one move. Do not merge two boxes.

Each topic page sets its own colors. The number in parentheses in the tables below is the topic.

prefixes

Symbol	Prefix	Power	Multiply by	In words
E	exa	10^{18}		quintillion
P	peta	10^{15}		quadrillion
T	tera	10^{12}	1,000,000,000,000	trillion
G	giga	10^{9}	1,000,000,000	billion
M	mega	10^{6}	1,000,000	million
k	kilo	10^{3}	1,000	thousand
—	(none)	10^{0}	1	one
m	milli	10^{-3}	0.001	thousandth
μ	micro	10^{-6}	0.000 001	millionth
n	nano	10^{-9}	0.000 000 001	billionth
p	pico	10^{-12}	0.000 000 000 001	trillionth
f	femto	10^{-15}		quadrillionth
a	atto	10^{-18}		quintillionth

Capital M (mega) and small m (milli) are 10^9 apart. [book Table 1-2]

quantities and units

Quantity	Symbol	Unit	In base units
charge	q, Q	coulomb (C)	A·s
current	i, I	ampere (A)	C/s
voltage	v, V (book: υ)	volt (V)	J/C
power	p, P	watt (W)	J/s = V·A
energy	w, W	joule (J)	W·s
resistance	R	ohm (Ω)	V/A
conductance	G	siemens (S)	A/V = $1/\Omega$
resistivity	ρ (rho)	ohm-meter	Ω·m
conductivity	σ (sigma)	siemens/meter	S/m
capacitance	C	farad (F)	C/V
inductance	L	henry (H)	V·s/A
frequency	f	hertz (Hz)	1/s
length, area	ℓ, A	m, m^2	—
time	t	second (s)	—
temperature	T	kelvin (K), °C	K = °C+273

Lowercase (q, i, v, p, w) can change with time. Uppercase (Q, I, V, P, W) is a constant (dc). [book Table 1-1]

constants and conversions

Item	Value
charge of one electron	$-e$, $e = 1.6 \times 10^{-19}$ C (more exact: 1.602)
charge of one proton	$+e$
electrons in 1 C	6.25×10^{18}
minute, hour	1 min = 60 s, 1 h = 3600 s
ampere-hour	1 Ah = 3600 C, 1 mAh = 3.6 C
watt-hour, kilowatt-hour	1 Wh = 3600 J, 1 kWh = 3.6×10^6 J
length	1 mm = 10^{-3} m, 1 cm = 10^{-2} m
area	1 mm^2 = 10^{-6} m^2, 1 cm^2 = 10^{-4} m^2
specific heat of water	$c = 4186$ J/(kg·°C)
mass of water	1 L ≈ 1 kg, 1 fl oz ≈ 0.02957 kg

The book uses $e = 1.6 \times 10^{-19}$ C. Use it to match the book's answers.

circuit terminology

Term	Meaning [book Table 1-4]
node	all the wire between elements: one voltage. Points joined by a wire or a short are the same node
ordinary node	connects to 2 elements
extraordinary node	connects to 3 or more elements
branch	one element between two nodes
path	a chain of branches that repeats no node
extraordinary path	a path between two adjacent extraordinary nodes
loop	a closed path
independent loop	a loop with at least one branch that no other chosen loop has
mesh	a loop that encloses no other loop
in series	same current: only ordinary nodes between them
in parallel	same voltage: they share both end nodes
planar circuit	can be drawn with no crossing wires. A crossing without a dot is not a connection
open circuit	$i = 0$, $R = \infty$, any voltage
short circuit	$v = 0$, $R = 0$, any current
ground	the 0 V reference node. Chassis ground: the metal frame. Earth ground: a rod in the soil

conductivity and resistivity, 20°C

Material	σ (S/m)	$\rho = 1/\sigma$ (Ω·m)
Conductors		
silver	6.17×10^7	1.62×10^{-8}
copper	5.81×10^7	1.72×10^{-8}
gold	4.10×10^7	2.44×10^{-8}
aluminum	3.82×10^7	2.62×10^{-8}
iron	1.03×10^7	9.71×10^{-8}
mercury (liquid)	1.04×10^6	9.58×10^{-7}
Semiconductors		
carbon (graphite)	7.14×10^4	1.40×10^{-5}
pure germanium	2.13	0.47
pure silicon	4.35×10^{-4}	2.30×10^3
Insulators		
paper	$\sim 10^{-10}$	$\sim 10^{10}$
glass	$\sim 10^{-12}$	$\sim 10^{12}$
teflon	$\sim 3.3 \times 10^{-13}$	$\sim 3 \times 10^{12}$
porcelain	$\sim 10^{-14}$	$\sim 10^{14}$
mica	$\sim 10^{-15}$	$\sim 10^{15}$
polystyrene	$\sim 10^{-16}$	$\sim 10^{16}$
fused quartz	$\sim 10^{-17}$	$\sim 10^{17}$
Common materials		
distilled water	5.5×10^{-6}	1.8×10^5
drinking water	$\sim 5 \times 10^{-3}$	~ 200
sea water	4.8	0.2
rubber	10^{-13}	10^{13}
Biological tissues		
blood, muscle	~ 1.5	~ 0.67
fat	~ 0.1	10

Small ρ (big σ) = good conductor. [book Table 2-1] If a problem gives ρ or σ, use the problem's value.

wire size (AWG) and cross-section area

AWG	Diameter d (mm)	Area $A = \pi d^2/4$ (m^2)
0	8.3	5.41×10^{-5}
2	6.5	3.32×10^{-5}
4	5.2	2.12×10^{-5}
6	4.1	1.32×10^{-5}
10	2.6	5.31×10^{-6}
14	1.6	2.01×10^{-6}
18	1.0	7.85×10^{-7}
20	0.8	5.03×10^{-7}

Bigger AWG number = thinner wire. [book Table 2-2]

Formula	In words
$A = \pi r^2 = \dfrac{\pi d^2}{4}$	round wire: (area) $= \pi$(radius)^2
$d = \sqrt{\dfrac{4A}{\pi}}$	(diameter) from the area
$A = w \cdot h$	rectangle: (width)(thickness), the face the current passes *through*
$A = \pi(b^2 - a^2)$	hollow tube: outer radius b, inner radius a

Convert mm to m before you square: $(1\text{ mm})^2 = 10^{-6}\text{ m}^2$. ℓ is along the current, A is across it. For a block, the current direction decides which side is ℓ.

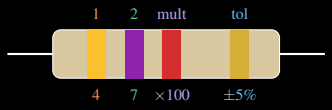
resistor color code

	Color	Digit	Multiplier	Tolerance	Temp. coef.
	black	0	$\times 1$	—	—
	brown	1	$\times 10$	$\pm 1\%$	100 ppm
	red	2	$\times 100$	$\pm 2\%$	50 ppm
	orange	3	$\times 1$k	—	15 ppm
	yellow	4	$\times 10$k	—	25 ppm
	green	5	$\times 100$k	$\pm 0.5\%$	—
	blue	6	$\times 1$M	$\pm 0.25\%$	10 ppm
	purple (violet)	7	$\times 10$M	$\pm 0.1\%$	5 ppm
	gray	8	—	—	—
	white	9	—	—	—
	gold	—	$\times 0.1$	$\pm 5\%$	—
	silver	—	$\times 0.01$	$\pm 10\%$	—

[book Fig. 2-2] Hold it so the wider gap is before the last band(s), on the right. Read left to right.

Formula	In words
4 bands: $R = (b_1 b_2) \times 10^{b_4} \pm b_5$	(two digits) $\times$ (multiplier) $\pm$ (tolerance)
5 bands: $R = (b_1 b_2 b_3) \times 10^{b_4} \pm b_5$	(three digits) $\times$ (multiplier) $\pm$ (tolerance)
6 bands: add b_6	5-band reading plus the temperature coefficient, ppm/°C

The multiplier band is the number of zeros. Digits 2–7 follow the rainbow: red, orange, yellow, green, blue, purple.



Steps — read the bands

1. ORIENT
Wide gap on the right. Read left to right.

↓

2. READ THE PIECES
digit 1, digit 2 (5 bands: a third digit), multiplier, tolerance

↓

3. TEMPLATE
$R = (\boxed{\text{digit 1}}\boxed{\text{digit 2}}) \times \boxed{\text{multiplier}} \pm \boxed{\text{tolerance}}$

↓

4. RANGE
$R \pm R \cdot \boxed{\text{tolerance}}$

↓

5. REVERSE ($R \to$ colors)
Write R as two digits $\times 10^k$. Bands: digit, digit, k zeros.

Quick example	Work → answer
brown, black, orange, gold	$10 \times 10^3 = 10\text{ k}\Omega$, $\pm 5\%$
5 bands: brown, black, black, red, brown	$100 \times 10^2 = 10\text{ k}\Omega$, $\pm 1\%$
red, red, gold, gold	$22 \times 0.1 = 2.2\,\Omega$, $\pm 5\%$
(reverse) $330\,\Omega$, $\pm 5\%$	33×10^1: orange, orange, brown, gold

formula sheet (symbols only)

2. charge, current $Q = It$, $i = \dfrac{dq}{dt}$, $\Delta Q = \int_{t_1}^{t_2} i\,dt$, $n = \dfrac{|Q|}{e}$

3. voltage $v_{ab} = \dfrac{dw}{dq}$, $V_{ab} = V_a - V_b$, $W = QV$

4. power, energy $p = vi$, $\sum p = 0$, $w = \int p\,dt$, $W = Pt$, $P_{av} = \dfrac{1}{T}\int_0^T p\,dt$

5. extras $\text{Wh} = V \cdot \text{Ah}$, $t = \dfrac{\text{Ah}}{I}$, $E = cm\Delta T$, $\text{kWh} = \dfrac{\text{J}}{3.6 \times 10^6}$

6. sources $V_L = \dfrac{R_L}{R_s + R_L} v_s$. dependent: αv_x, $r i_x$, $g v_x$, βi_x

7. resistors $V = IR$, $R = \dfrac{\rho \ell}{A} = \dfrac{\ell}{\sigma A}$, $P = I^2R = \dfrac{V^2}{R}$, $R = R_0(1 + \alpha T)$

8. KCL, KVL $\sum i_{\text{in}} = \sum i_{\text{out}}$, $\sum v = 0$. N unknowns: M KVL + $(N - M)$ KCL

9. combine $R_1 + R_2$, $\dfrac{R_1 R_2}{R_1 + R_2}$, $G_{eq} = \sum G$

10. dividers $v_1 = \dfrac{R_1}{R_1 + R_2} v_s$, $i_1 = \dfrac{R_2}{R_1 + R_2} i_s = \dfrac{G_1}{G_{eq}} i_s$

11. source transf. $i_s = \dfrac{v_s}{R_s}$, $v_s = i_s R_s$ (same R_s)

12. $\Delta \to Y$ $R_1 = \dfrac{R_b R_c}{R_a + R_b + R_c}$, $R_a = \dfrac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_1}$

13. bridge $R_x = \dfrac{R_2}{R_1} R_3$, $V_{out} \approx \dfrac{V_0}{4} \dfrac{\Delta R}{R}$, $R = R_0(1 + \alpha P)$

14. diode, LED forward: $I = \dfrac{V_s - V_F}{R}$, $V_F \approx 0.7$ V (Si)

16. linearity sources $\times K$: $v, i \times K$, $p \times K^2$. unit output: $K = \dfrac{v_s}{v_s'}$

17. mesh $(\sum R_k) I_k - \sum R_{kj} I_j = \sum v_{\text{rise}}$, $i_{\text{shared}} = I_a - I_b$, supermesh: $I_b - I_a = i_s$

18. superposition $v = v_1 + \cdots + v_n$. off: $v_s \to$ short, $i_s \to$ open. $p \neq \sum p_k$

19. validation $\%\,\text{err} = \dfrac{|\text{sim} - \text{hand}|}{|\text{hand}|} 100$

calculus and algebra tools

Derivatives (a, ω, n, c constants):

Function	Derivative
c, t^n	0, nt^{n-1}
e^{at}	$a e^{at}$
$\sin(\omega t)$, $\cos(\omega t)$	$\omega \cos(\omega t)$, $-\omega \sin(\omega t)$
$f \cdot g$ (product)	$f'g + fg'$
$f(g(t))$ (chain)	$f'(g) \cdot g'$. For example, $\sin^2(\omega t) \to 2\omega \sin(\omega t)\cos(\omega t)$

Integrals:

Function	Antiderivative
t^n	$\frac{t^{n+1}}{n+1}$
e^{at}	$\frac{e^{at}}{a}$
$\sin(\omega t),\quad \cos(\omega t)$	$-\frac{\cos(\omega t)}{\omega},\quad \frac{\sin(\omega t)}{\omega}$
$e^{at}\cos(bt)$	$\frac{e^{at}\left(a\cos bt + b\sin bt\right)}{a^2+b^2}$
$e^{at}\sin(bt)$	$\frac{e^{at}\left(a\sin bt - b\cos bt\right)}{a^2+b^2}$
$\int_0^\infty Ae^{-at}\,dt$	$\frac{A}{a}\quad (a>0)$

Definite integral: $F(\text{top}) - F(\text{bottom})$. $\quad e^0 = 1$. $\quad e^{-at} \to 0$ as $t \to \infty$.
Trig identities: $\sin x\cos x = \frac{1}{2}\sin 2x$, $\quad \cos^2 x = \frac{1}{2}(1+\cos 2x)$, $\sin^2 x = \frac{1}{2}(1-\cos 2x)$.
Maximum or minimum: set the derivative to 0 and solve for t. Example: largest q where $\frac{dq}{dt} = 0$, and the largest p where $\frac{dp}{dt} = 0$.

Solving simultaneous equations

2 unknowns ($a_1x + b_1y = c_1$, $\quad a_2x + b_2y = c_2$), Cramer's rule:

$$D = a_1b_2 - a_2b_1 \qquad x = \frac{c_1b_2 - c_2b_1}{D} \qquad y = \frac{a_1c_2 - a_2c_1}{D}$$

3 unknowns:

$$D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2).$$

$x_k = \frac{D_k}{D}$, where D_k is D with column k replaced by the right-hand sides. Or: eliminate one unknown at a time, or use a calculator's matrix solver (rref).
[book App. B]

Quick example	Work → answer
$3x+2y=12,\quad x-y=-1$	$D = 3(-1) - 1(2) = -5.\quad x = \frac{12(-1)-(-1)(2)}{-5} = 2,$ $y = \frac{3(-1)-1(12)}{-5} = 3$

which tool? — given → find

The number in parentheses is the topic.

Given → find	Do this	
a circuit → nodes, series, parallel	terminology table	(1)
charge Q → electrons	$n = \lvert Q\rvert/e$	(2)
$q(t) \to i$ / $i(t) \to \Delta Q$	differentiate / integrate	(2)
a $q(t)$ or $i(t)$ graph	slope / area	(2)
node voltages, new ground	subtract the new ground's old value	(3)
V, I → power, sign	$P = VI$, passive sign convention	(4)
$v(t)$, $i(t)$ → energy, average power	$\int vi\,dt$ / $\frac{1}{T}\int p\,dt$	(4)
battery, kWh, heat, cost	extras	(5)
switch at t_0	redraw: closed = short, open = open	(6)
two of V, I, R → the third	Ohm's law	(7)
wire: $R \leftrightarrow \ell$, A, material	$R = \rho\ell/A$ and the tables	(7)
resistance at a new temperature	$R = R_0(1+\alpha T)$	(7)
color bands → R	color chart (ref)	
currents at a node / voltages in a loop	KCL / KVL	(8)
N unknown currents	M KVL + $(N-M)$ KCL, solve	(8)
network → R_{eq}	combination chart	(9)
series → v share / parallel → i share	dividers	(10)
mixed sources and resistors	source transformation	(11)
not series, not parallel (a bridge)	$\Delta \leftrightarrow Y$	(12)
bridge balance / sensor output	Wheatstone	(13)
force on a sensor → V_{out}	piezoresistor, cantilever	(13)
diode or LED in a circuit	assume on, check $I > 0$	(14)
a known answer, new source value	scale by K, power by K^2	(16)
a ladder → output	unit output, then scale	(16)
planar circuit, many loops	mesh currents, one KVL per mesh	(17)
current source between two meshes	supermesh	(17)
several independent sources	superposition, one at a time	(18)
simulation vs hand answer	percent error, same ground and directions	(19)

1–3 — reading a circuit, charge and current, voltage

1. Reading a circuit [book 1-3]

EXPLANATION

THE IDEA
Label the nodes first. Every wire segment between elements is one node, even if it is drawn long.

Pieces

Piece	Where to find it
ordinary node	a node that touches exactly 2 elements
extraordinary node	a node that touches 3 or more
branch	one element (or a series chain) between two nodes
mesh	a loop with no other loop inside

Meters [book Fig. 1-18]: voltmeter in parallel (across), red lead on $+$. Ammeter in series (in the path).
Trap: Points joined by a plain wire are the same node. Merge them.
Trap: Same branch, same direction $=$ same current. Opposite arrows are negatives.

WORK

1. LABEL THE NODES
Mark every wire segment between elements. Merge wire-joined points.

↓

2. WHAT TOUCHES EACH NODE
List the elements at each node. $2 \rightarrow$ ordinary. $3+ \rightarrow$ extraordinary.

↓

3. SERIES?
Only an [ordinary node] between two elements $\Rightarrow$ series.

↓

4. PARALLEL?
Two elements share [both end nodes] $\Rightarrow$ parallel.

↓

5. COUNT MESHES
[meshes] $=$ [branches] $-$ nodes $+ 1$ *(check against the drawing)*

2. Charge and current [book 1-4]

EXPLANATION

THE IDEA
Current is the flow of positive charge. Electrons moving one way count as current the other way.

Pieces

Piece	Where to find it
$q(t)$	charge as a function of time
$i(t)$	current as a function of time
t_1, t_2	the time window (seconds)
Q	a total charge. $e = 1.6 \times 10^{-19}$ C

Formula	In words
$n = \dfrac{\lvert Q \rvert}{e}$	(electrons) $=$ (charge) $\div$ (charge of one)
$Q = It, \quad I = \dfrac{Q}{t}, \quad t = \dfrac{Q}{I}$	constant current
$i = \dfrac{dq}{dt}$	differentiate the charge
$\Delta Q = \int_{t_1}^{t_2} i\,dt$	integrate the current
$\int_0^\infty Ae^{-at}\,dt = \dfrac{A}{a}$	shortcut, $a > 0$

Trap: Given charge, find current: differentiate. A higher power of t means you integrated.
Trap: Electrons one way plus protons the other way: both count toward the same current direction.
Trap: Time in seconds.

WORK

Steps — $q(t) \rightarrow i(t)$

1. DIFFERENTIATE
$i = \dfrac{d}{dt}[q(t)]$ *(product rule: $(fg)' = f'g + fg'$)*

Steps — $i(t) \rightarrow \Delta Q$

1. ANTIDERIVATIVE
$F(t) = \int [i(t)]\,dt$

↓

2. TOP MINUS BOTTOM
$\Delta Q = F([t_2]) - F([t_1])$ *($e^0 = 1$)*

Steps — from a graph

1. q GRAPH $\rightarrow$ CURRENT
current $=$ slope of each straight piece of q

↓

2. i GRAPH $\rightarrow$ CHARGE
charge $=$ area under i, piece by piece

Steps — largest charge

1. SET $\frac{dq}{dt} = 0$
Solve for t. Plug that t back into $q(t)$.

3. Voltage, node voltage, ground [book 1-5.1]

EXPLANATION

THE IDEA
Voltage is energy per charge between two points. A **node voltage** is measured from ground (0 V).

Formula	In words
$V = \dfrac{W}{Q}, \quad W = QV$	energy per charge
$V_{ab} = V_a - V_b$	between two nodes
$V_{ab} = -V_{ba}$	reverse order flips the sign

Trap: Moving ground shifts every node voltage. Differences do not change.

WORK

Steps — move the ground

1. OLD VALUE AT THE NEW GROUND
Read $V_{\text{new ground}}$ in the old labels.

↓

2. SUBTRACT IT FROM EVERY NODE
$V_{\text{new}} = V_{\text{old}} - V_{\text{new ground}}$

↓

3. CHECK
Any V_{ab} is the same before and after.

4–5 — power and energy

4. Power and the passive sign convention [book 1-5.2]

EXPLANATION

THE IDEA
Power is $v \cdot i$. The sign tells you whether the device receives power or supplies it.

Pieces

Piece	Where to find it
$+$ side	the current arrow enters the $+$ mark of v
$-$ side	the current arrow enters the $-$ mark
v, i	the values, with their own signs

Formula	In words
$p = vi, \quad I = \dfrac{P}{V}, \quad V = \dfrac{P}{I}$	power
$\sum p_k = 0$	supplied = received
$W = Pt, \quad W = VIt$	constant power
$w = \int_{t_1}^{t_2} v\,i\,dt$	energy from $v(t)$, $i(t)$
$P_{av} = \dfrac{1}{T}\int_0^T p\,dt$	average power

Trap: $\int v\,i\,dt \neq \int v\,dt \cdot \int i\,dt$. Multiply first.
Trap: A resistor always receives power.

WORK

Steps — sign of the power

1. WHICH SIDE?
Does i enter the $+$ or the $-$ of v?

↓

2. TEMPLATE
$p = \boxed{+}/\boxed{-}\, v\,i$

↓

3. READ
$p > 0$: receives. $\quad p < 0$: supplies.

↓

4. CHECK
Every p in the circuit adds to 0.

Steps — energy from $v(t)$ and $i(t)$

1. MULTIPLY FIRST
$p(t) = v(t) \cdot i(t)$. Simplify: $e^{-at}e^{-bt} = e^{-(a+b)t}$.

↓

2. INTEGRATE
$w = \displaystyle\int_{t_1}^{t_2} p\,dt$. To ∞: $\int_0^\infty Ae^{-at}\,dt = \frac{A}{a}$ per term.

↓

3. UNITS
V·mA·s = mJ.

Steps — maximum power

1. SET $\frac{dp}{dt} = 0$
Solve for t, then plug into $p(t)$.

5. Batteries, utility energy, heat, cost

EXPLANATION

THE IDEA
Ah is charge. Wh and kWh are energy. Convert everything to joules or watt-hours before you compare.

Formula	In words
$\text{Wh} = V \cdot \text{Ah}$	energy rating
$t = \dfrac{\text{Ah}}{I} = \dfrac{\text{Wh}}{P}$	run time, hours
$\text{J} = \text{Wh} \cdot 3600$	hours to seconds
$E = cm\Delta T, \quad \Delta T = \dfrac{E}{cm}$	heat (water: $c = 4186$)
$\text{cost} = \dfrac{\text{J}}{3.6 \times 10^6} \cdot \text{rate}$	joules → kWh → cost

Trap: A count of batteries filled rounds down.

WORK

Steps — run time

1. SAME UNITS
mAh → Ah, mA → A.

↓

2. DIVIDE
$t = \dfrac{\text{Ah}}{I}$ or $t = \dfrac{\text{Wh}}{P}$ (hours)

Steps — how many batteries

1. ONE BATTERY IN J
$W_{\text{one}} = V \cdot \text{Ah} \cdot 3600$

↓

2. DIVIDE, ROUND DOWN
$n = \dfrac{W}{W_{\text{one}}}$

Steps — heat and cost

1. HEAT
$E = cm\Delta T$ (J)

↓

2. TO kWh
$\dfrac{E}{3.6 \times 10^6}$

↓

3. COST
kWh × rate

6–7 — sources, switches, Ohm's law, resistance

6. Sources and switches [book 1-6]

EXPLANATION

THE IDEA
An ideal voltage source fixes v for any current. An ideal current source fixes i for any voltage. A real source adds an internal R_s.

Element	Rule
ideal voltage source	$v = V_s$, any i
ideal current source	$i = I_s$, any v
real voltage source	v_s in series with R_s
real current source	i_s in parallel with R_s
switch	closed = short, open = open, changes at t_0

Dependent source	Value, gain unit
VCVS	αv_x, no unit
CCVS	$r i_x$, Ω
VCCS	$g v_x$, S
CCCS	βi_x, no unit

Trap: A dependent source is a model, not an energy source. Its value comes from v_x or i_x.

WORK

Steps — load voltage of a real source

1. DIVIDER
$$V_L = \frac{\boxed{R_L}}{\boxed{R_s}+\boxed{R_L}}\, v_s$$

Steps — a dependent source

1. FIND THE CONTROL
Write v_x or i_x in terms of your unknowns.

↓

2. VALUE
gain × control: αv_x, $r i_x$, $g v_x$, βi_x

↓

3. TREAT AS A SOURCE
Use it in KVL or KCL like any source.

Steps — a switch

1. TWO CIRCUITS
Draw $t < t_0$ and $t > t_0$. Closed = short, open = open.

7. Ohm's law, resistance of a shape, power rating [book 2-1]

EXPLANATION

THE IDEA
$V = IR$ for one resistor. The shape and material set R. The power rating caps the current.

Pieces

Piece	Where to find it
ρ or σ	the material table (Table 2-1)
ℓ	length. A two-wire cable: twice the run
A	cross-section area. Wire: $\frac{\pi d^2}{4}$, d from AWG
R_0, α	resistance at 0°C and the temperature coefficient

Formula	In words
$V = IR,\quad I = \frac{V}{R},\quad R = \frac{V}{I}$	Ohm's law
$I = \frac{V_a - V_b}{R}$	current a → b
$R = \frac{\rho\ell}{A},\quad \ell = \frac{RA}{\rho},\quad A = \frac{\rho\ell}{R}$	shape
$R = R_0(1+\alpha T)$, $T = \frac{R/R_0 - 1}{\alpha}$	temperature
$P = I^2R = \frac{V^2}{R}$	power in a resistor
$I_{max} = \sqrt{\frac{P_{rated}}{R}}$, $V_{max} = \sqrt{P_{rated}R}$	safe limits

Trap: If I enters the − side, $I = -V/R$.
Trap: Convert mm to m before squaring.

WORK

Steps — wire design ($R \to \ell$)

1. R
$R = \frac{V}{I}$ if not given.

↓

2. DIAMETER
d from the AWG table, in meters.

↓

3. AREA
$$\boxed{A} = \frac{\pi d^2}{4}$$

↓

4. LENGTH
$$\boxed{\ell} = \frac{R\boxed{A}}{\boxed{\rho}}$$ *(a cable run is $\ell/2$)*

Steps — resistance at a temperature

1. TEMPLATE
$R = \boxed{R_0}\left(1+\boxed{\alpha}T\right)$ *(T for a target R: $\frac{R/R_0-1}{\alpha}$)*

Steps — power rating

1. LIMIT
$I_{max} = \sqrt{P_{rated}/R}$. Your I above it ⇒ too small a rating.

↓

2. FRACTION TO THE LOAD
$$\frac{I^2R_L}{V_sI}$$

8 — Kirchhoff's laws and nodal analysis [book 2-2, 3-2]

KCL, KVL, and the path rule

EXPLANATION

THE IDEA
KCL: what flows into a node flows out. KVL: around any closed loop the voltages add to 0.

Pieces

Piece	Where to find it
in	arrows pointing into the node
out	arrows pointing out of the node
+ first	walking the loop, you meet the + mark first: write $+v$
− first	you meet the − mark first: write $-v$

Formula	In words
$\sum i_{\text{in}} = \sum i_{\text{out}}$	KCL
$\sum v = 0$ around a loop	KVL
V_{ab} = sum along any path a → b	path rule: $+v$ if you meet + first

Polarity on a resistor: put + where your current arrow enters. Then $v = +iR$.
Trap: A negative answer is fine: the current flows opposite to your arrow.
Trap: Never write KVL through a current source. Its voltage is unknown.

WORK

Steps — KCL at a node

1. TEMPLATE
$\sum$ [in] = $\sum$ [out]. Solve for the unknown.

Steps — KVL around a loop

1. MARK POLARITIES
Each resistor: + where its current arrow enters. Each source: as drawn.

↓

2. WALK CLOCKWISE
Start at any corner. One term per element: [$+v$] if + first, [$-v$] if − first.

↓

3. OHM IN
Each resistor term: $v = iR$.

↓

4. SET TO ZERO, SOLVE
$\sum = 0$.

Steps — voltage between two points (path rule)

1. WALK a → b
Any path. $+v$ if you meet + first, $-v$ if − first. The sum is V_{ab}.

↓

2. CHECK
Walk another path. Same answer.

Branch-current method: KCL and KVL together [book 2-2.3]

EXPLANATION

THE IDEA
Every branch gets its own unknown current. KCL and KVL together give exactly as many equations as unknowns.

Pieces

Piece	Where to find it
N	the number of unknown branch currents
n_{ex}	extraordinary nodes (3 or more elements)
M	meshes that do not pass through a current source
known currents	a branch with a current source: that current is given

Formula	In words
KCL equations = $n_{ex} - 1$	one node is left out: it repeats the others
KVL equations = $N - (n_{ex} - 1)$	the rest, one per mesh

Trap: Each new KVL loop must use an element not used before.
Trap: Arrow directions are guesses. Keep them. A negative answer means reversed.

WORK

1. LABEL
An arrow and a name $i_1, i_2, \ldots$ on each branch. Known source currents too.

↓

2. POLARITIES
+ where each arrow enters each resistor.

↓

3. COUNT
N unknowns. Need [n_{ex}] − 1 KCL and the rest KVL.

↓

4. KCL
At [n_{ex}] − 1 extraordinary nodes: in = out.

↓

5. KVL
Around meshes that skip current sources. Resistor terms as iR.

↓

6. SOLVE THE SYSTEM
See "Solving the system" below: solve one, plug back in.

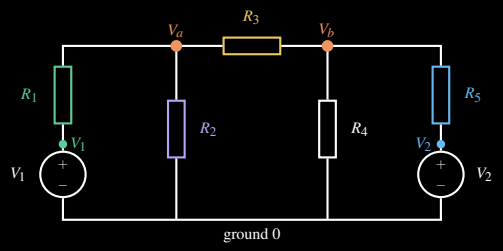
↓

7. CHECK
The unused KCL node balances. Power: supplied = received.

Nodal analysis: the walk [book 3-2]

EXPLANATION

THE IDEA
Unknowns are node voltages. Stand at an unknown node, cross each resistor, and note where you land. One KCL per unknown node.



Pieces

Piece	Where to find it
V_{here}	the unknown node you stand at (V_a, then V_b)
far node	the node across each R: a known value, 0, or another unknown
known node	a source top: $+v_s$, $-v_s$ if $-$ is on top
shared R	joins two unknowns. In both equations, flipped

At V_a above	Fractions
through R_1 to V_1	$\frac{V_a - V_1}{R_1}$
through R_2 to ground	$\frac{V_a - 0}{R_2}$
through R_3 to V_b	$\frac{V_a - V_b}{R_3}$

Trap: Multiply *every* term by the LCD. A minus before parentheses flips every term inside.
Trap: At V_b the shared term is $\frac{V_b - V_a}{R_3}$: here minus far.

WORK

1. GROUND
Pick the node most sources touch. It is 0 V.

2. KNOWN NODES
Each source top is known: $\pm v_s$. *(voltage source between two unknowns: supernode, below)*

3. LIST THE UNKNOWNS
$V_a, V_b, \ldots$ One equation each.

4. KCL AT EACH UNKNOWN
$$\sum \frac{\boxed{V_{\text{here}}} - \boxed{\text{far}}}{\boxed{R}} = 0$$ *(one fraction per R, all leaving. Current source leaving: $+i_s$, entering: $-i_s$)*

5. CLEAR FRACTIONS
Multiply every term by the LCD of that equation.

6. STANDARD FORM
Collect: $(\ )V_a + (\ )V_b =$ number. Numbers to the right side.

7. SOLVE THE SYSTEM
One unknown: divide. Two or more: "Solving the system" below.

8. PLUG BACK IN
Each V into every original equation: each must give 0.

9. CURRENTS AND POWER
$i = \frac{V_{\text{here}} - V_{\text{far}}}{R}$ for each R. $p = i^2 R$. Source: $p = v_s\, i$.

10. CHECK
KCL at each node with the numbers. Supplied power = received.

Supernode and dependent sources in nodal

EXPLANATION

THE IDEA
A voltage source between two unknown nodes hides its current. Put a boundary around both nodes and treat them as one big node.

Pieces

Piece	Where to find it
V_+, V_-	the nodes at the $+$ and $-$ ends of the floating source
v_s	the floating source
boundary	a loop drawn around both nodes and the source

Trap: Inside the boundary, skip the source branch. Its current is unknown.
Trap: A dependent source: write its control v_x or i_x in node voltages, for example $i_x = \frac{V_a - V_b}{R}$.

WORK

Steps — supernode

1. CONSTRAINT
$\boxed{V_+} - \boxed{V_-} = \boxed{v_s}$

2. ONE KCL FOR THE BOUNDARY
Sum every current leaving the boundary $= 0$. Each R leaving uses the node it starts from.

3. OTHER NODES
Normal KCL at any unknown node outside the boundary.

4. SOLVE
Substitute the constraint, then "Solving the system".

Steps — dependent source

1. CONTROL IN NODE VOLTAGES
$v_x = V_{\text{its}+} - V_{\text{its}-}$, $i_x = \frac{V_{\text{start}} - V_{\text{end}}}{R_x}$

2. SUBSTITUTE
Put that into the source value, then into KCL. Move unknowns left.

Solving the system (2 or 3 unknowns) [book App. B]

EXPLANATION

THE IDEA
Solve for one unknown, then plug it back in to get the rest. Every answer must satisfy every original equation.

Pieces (standard form)

Piece	Where to find it
a_1, b_1	coefficients of x, y in equation 1
a_2, b_2	coefficients of x, y in equation 2
c_1, c_2	the right-side numbers

$a_1x+b_1y=c_1 \qquad a_2x+b_2y=c_2$

If	Use
a coefficient is 1 or -1	substitution
coefficients are messy	Cramer's rule
3 unknowns	eliminate one to get 2, or 3×3 Cramer, or a calculator (rref)

Trap: Plug back into the *original* equations, not the rearranged ones.
Trap: Keep fractions exact until the end.

WORK

Steps — substitution

1. ISOLATE
Pick a variable with coefficient ±1. Solve its equation for it: $x=\ldots y\ldots$

↓

2. SUBSTITUTE
Put that into the *other* equation. Now one unknown.

↓

3. SOLVE
Find y.

↓

4. PLUG BACK IN
y into step 1: find x.

↓

5. CHECK
Both values into both original equations.

Steps — Cramer's rule (2 unknowns)

1. DETERMINANT
$D=\boxed{a_1}\boxed{b_2}-\boxed{a_2}\boxed{b_1}$ *($D=0$: no single answer)*

↓

2. FIRST UNKNOWN
$x=\dfrac{\boxed{c_1}\boxed{b_2}-\boxed{c_2}\boxed{b_1}}{D}$ *(replace the x column with c)*

↓

3. SECOND UNKNOWN
$y=\dfrac{\boxed{a_1}\boxed{c_2}-\boxed{a_2}\boxed{c_1}}{D}$ *(or plug x back in)*

Steps — 3 unknowns by elimination

1. ELIMINATE ONE
Combine equations 1 and 2, then 1 and 3, to cancel the same variable.

↓

2. SOLVE THE 2 BY 2
Substitution or Cramer.

↓

3. PLUG BACK IN
Both answers into any original equation for the third.

9–10 — series, parallel, and dividers [book 2-3]

9. Reduce a network to R_{eq}

EXPLANATION

THE IDEA
Replace a clear series pair or parallel pair with one resistor. Redraw. Repeat from the end farthest from the terminals.

Formula	In words
$R_1 + R_2 + \cdots$	series: add
$\dfrac{R_1 R_2}{R_1 + R_2}$	two in parallel: product over sum
$\dfrac{1}{1/R_1 + 1/R_2 + \cdots}$	any number in parallel
$\dfrac{R}{N}$	N equal in parallel
$R \parallel 0 = 0, \quad R \parallel \infty = R$	short across, open branch
$R = \dfrac{R_{eq} R_1}{R_1 - R_{eq}}$	missing R in parallel

Trap: Series total $>$ the biggest. Parallel total $<$ the smallest.

WORK

1. LABEL NODES
Merge nodes joined by wires or shorts.

↓

2. FARTHEST PAIR
Find a series or parallel pair far from the terminals a, b.

↓

3. REPLACE
series: $\boxed{R} + \boxed{R}$ parallel: $\dfrac{\boxed{R} \cdot \boxed{R}}{\boxed{R} + \boxed{R}}$

↓

4. REDRAW, REPEAT
Until one resistor is left between a and b.

↓

5. CHECK
Each parallel result is below its smaller part.

10. Voltage and current dividers

EXPLANATION

THE IDEA
Series resistors share the voltage, in proportion to R. Parallel resistors share the current, in proportion to $\frac{1}{R}$.

Pieces

Piece	Where to find it
own R	the resistor you want v or i for
other R	current divider: the other branch
every series R	voltage divider: the whole chain
source	v_s across the chain, or i_s into the pair

Formula	In words
$V_{\text{node}} = \dfrac{R_{\text{below}}}{R_T} v_s$	node voltage in a series stack
$i_k = \dfrac{R_{eq}}{R_k} i_s$	current divider, any number of branches
$R_2 = \dfrac{R_1 v_2}{v_s - v_2}$	design: R_2 for a target v_2
$R_2 = \dfrac{R_1 (i_s - i_2)}{i_2}$	design: R_2 for a target i_2

Trap: Voltage divider: own R on top. Current divider: other R on top.
Trap: A load across a divider changes it. Combine the load first.
Trap: Node voltage = voltage across only when the other end is ground.

WORK

Steps — voltage divider

1. TEMPLATE
$v = \dfrac{\boxed{\text{own } R}}{\boxed{\text{every series } R}} \cdot \boxed{v_s}$

↓

2. CHECK
All drops add to v_s.

Steps — current divider (two branches)

1. TEMPLATE
$i = \dfrac{\boxed{\text{other } R}}{\boxed{\text{own}} + \boxed{\text{other}}} \cdot \boxed{i_s}$

↓

2. CHECK
Branch currents add to i_s. Same voltage: $i_1 R_1 = i_2 R_2$.

Steps — loaded divider

1. MERGE THE LOAD
$\boxed{R_2 \parallel R_L}$ becomes one piece.

↓

2. DIVIDE
$v_{out} = \dfrac{\boxed{R_2 \parallel R_L}}{\boxed{R_1} + \boxed{R_2 \parallel R_L}} \boxed{v_s}$ *(lower than unloaded)*

11–12 — source transformation, wye–delta

11. Source transformation and combining sources [book 2-3]

EXPLANATION

THE IDEA

v_s in series with R_s acts exactly like i_s in parallel with the same R_s.

Pieces

Piece	Where to find it
v_s	the voltage source, with its $+$ end
i_s	the current source, with its arrow
R_s	the resistor in series (v_s) or parallel (i_s)

Formula	In words
$v_{eq} = v_1 - v_2 + \cdots$	voltage sources in series: add with signs
$i_{eq} = i_1 - i_2 + \cdots$	current sources in parallel: add with directions

Trap: Do not transform the element whose v or i you need.
Trap: Different voltage sources in parallel, or different current sources in series: unrealizable.

WORK

Steps — v_s series $\rightarrow$ i_s parallel

1. CURRENT

$$\boxed{i_s} = \frac{\boxed{v_s}}{\boxed{R_s}}$$

↓

2. REDRAW

Same R_s, now in parallel. The arrow points toward the old $+$.

Steps — i_s parallel $\rightarrow$ v_s series

1. VOLTAGE

$$\boxed{v_s} = \boxed{i_s}\,\boxed{R_s}$$

↓

2. REDRAW

Same R_s, now in series. $+$ at the end the arrow pointed to.

↓

3. CHECK

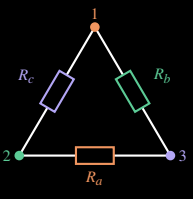
Open terminals: same V_{oc}. Shorted: same I_{sc}.

12. Wye–delta (Y–Δ) [book 2-4]

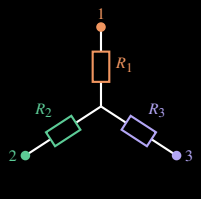
EXPLANATION

THE IDEA

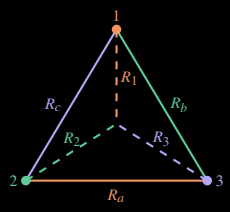
For a network that is neither series nor parallel (a bridge). Both shapes connect the same three nodes 1, 2, 3. Each node has one color: the Y arm *at* it and the Δ side *opposite* it share that color.



Δ: side *opposite* its node



Y: arm *at* its node



Overlay: arm R_1 sits between the sides that touch node 1, R_b and R_c. It points at its own-color side R_a.

Pieces (book labels)

Piece	Where to find it
R_1, R_2, R_3	Y arms, at nodes 1, 2, 3
R_a, R_b, R_c	Δ sides, opposite nodes 1, 2, 3
Σ	$R_a + R_b + R_c$
N	$R_1R_2 + R_2R_3 + R_1R_3$

Color rule	Formula
Δ → Y: arm = the other two colors	$R_1 = \dfrac{R_bR_c}{\Sigma}$
Y → Δ: side = $N \div$ the same color	$R_a = \dfrac{N}{R_1}$
balanced (all equal)	$R_Y = \frac{R_\Delta}{3}$, $R_\Delta = 3R_Y$

Trap: Other books swap the letters. Match by position: an arm uses the two sides that *touch* its node.

WORK

Steps — spot it in a circuit

1. LOOK FOR A TRIANGLE OR A STAR

Δ: three resistors in a closed triangle. Y: three resistors meeting at one node that touches nothing else.

↓

2. NAME THE THREE NODES

Call them 1, 2, 3. They stay in the circuit after the change.

↓

3. COLOR EACH RESISTOR

Y arm: the color of its node. Δ side: the color of the node it does *not* touch.

Steps — Δ → Y

1. SUM

$\Sigma = \boxed{R_a} + \boxed{R_b} + \boxed{R_c}$

↓

2. EACH ARM

$\boxed{R_1} = \dfrac{\boxed{R_b}\,\boxed{R_c}}{\Sigma}$ $\boxed{R_2} = \dfrac{\boxed{R_a}\,\boxed{R_c}}{\Sigma}$ $\boxed{R_3} = \dfrac{\boxed{R_a}\,\boxed{R_b}}{\Sigma}$

↓

3. CHECK

Nodes 1–2, node 3 open: Y: $R_1 + R_2$ = Δ: $R_c \parallel (R_a + R_b)$

Steps — Y → Δ

1. PAIR PRODUCTS

$N = \boxed{R_1}\,\boxed{R_2} + \boxed{R_2}\,\boxed{R_3} + \boxed{R_1}\,\boxed{R_3}$

↓

2. EACH SIDE

$\boxed{R_a} = \dfrac{N}{\boxed{R_1}}$ $\boxed{R_b} = \dfrac{N}{\boxed{R_2}}$ $\boxed{R_c} = \dfrac{N}{\boxed{R_3}}$

↓

3. CHECK

Convert back. You get the arms you started with.

Steps — finish the circuit

1. REDRAW

Put the new shape on the same nodes 1, 2, 3.

↓

2. REDUCE

Now series and parallel pairs appear. Use §9.

13–14 — Wheatstone bridge, sensors, nonlinear elements

13. Wheatstone bridge and sensors [book 2-5, 2-6.3]

EXPLANATION

THE IDEA
Two voltage dividers side by side. Balanced means both middles sit at the same voltage, so no current crosses.

Pieces

Piece	Where to find it
tops	R_1 (left), R_2 (right)
bottoms	R_3 (left), R_x (right)
V_0	the bridge supply
$\Delta R/R$	the sensor's fractional change

Formula	In words
$V_{out} = \dfrac{V_0 \Delta R}{4R + 2\Delta R}$	exact, one arm changed
$R = R_0(1 + \alpha P)$	piezoresistor, P in N/m^2
$P = \dfrac{F}{A}$, cantilever $P = \dfrac{FL}{WH^2}$	pressure

Silicon: $\alpha \approx -10^{-9}$ m^2/N.

WORK

Steps — balance

1. SAME RATIO EACH SIDE

$$\frac{\boxed{\text{left top}}}{\boxed{\text{left bottom}}} = \frac{\boxed{\text{right top}}}{\boxed{\text{right bottom}}}$$

↓

2. SOLVE

$$R_x = \frac{R_2}{R_1} R_3$$

Steps — sensor output

1. SMALL CHANGE

$$V_{out} \approx \frac{\boxed{V_0}}{4} \cdot \boxed{\Delta R/R}$$

↓

2. REVERSE

$$\frac{\Delta R}{R} = \frac{4 V_{out}}{V_0}$$

Steps — piezoresistor chain

1. PRESSURE

$$P = \frac{F}{A} \text{ or } \frac{FL}{WH^2}$$

↓

2. CHANGE

$$\frac{\Delta R}{R} = \alpha P$$

↓

3. OUTPUT

$$V_{out} = \frac{V_0}{4} \alpha P$$

14. Nonlinear elements: fuse, diode, LED [book 2-1.4, 2-6]

EXPLANATION

THE IDEA
A diode is a one-way valve. Off: an open. On: a fixed drop V_F.

Element	Rule
fuse	opens above its rated current
ideal diode	forward: short, reverse: open
real diode	on at $V_F \approx 0.7$ V (Si)
LED	red $V_F \approx 1.6$ V, green or blue ≈ 3.3 V, about 20 mA

Formula	In words
$R_D \approx \dfrac{\Delta V_D}{\Delta I}$	slope of the curve above V_F

Trap: Superposition and scaling do not work with diodes (§16).

WORK

Steps — diode in a circuit

1. ASSUME ON
Replace it with a V_F drop.

↓

2. CURRENT

$$I = \frac{V_s - V_F}{R}$$

↓

3. CHECK THE GUESS
$I > 0$: on, done. $I \le 0$: off, $I = 0$.

Steps — LED resistor

1. TEMPLATE

$$R = \frac{V_s - V_F}{I}$$

15–16 — one full circuit, linear circuits

15. Putting it together: reduce, then expand

EXPLANATION

THE IDEA
Reduce to one resistor and find the source current. Then walk back out with Ohm's law, KVL and KCL.

Trap: Write down each reduced circuit. You walk back through them in reverse.

WORK

1. REDUCE
R_{eq} with §9.

↓

2. SOURCE CURRENT
$i = \dfrac{v_s}{R_{eq}}$

↓

3. DROPS
$v = iR$ on each series piece. KVL for what is left.

↓

4. BRANCH CURRENTS
$i = \dfrac{v}{R}$ in each parallel branch. KCL check.

↓

5. CHECK (POWER)
$\sum i^2 R = v_s\, i$

16. Linear circuits [book 3-1]

EXPLANATION

THE IDEA
A **linear circuit** has only independent sources and linear elements. Scale every source by K and every v and i scales by K.

Pieces

Piece	Where to find it
old source	the source value of the answer you already have
new source	the changed source value
known answer	any v, i or p from the old circuit

Formula	In words
$v = R(Ki) = K(Ri)$	**homogeneity**: scale in, scale out
$v = R(i_1 + i_2) = Ri_1 + Ri_2$	**additivity**: add in, add out
sources $\times K$	every v, $i \times K$, every $p \times K^2$

Linear	Not linear
R, C, L	diode, LED, fuse
dependent source, gain $\times$ one v or i	depends on v_x^2 or $v_x i_y$

Trap: Power is not linear: $2\times$ the source gives $4\times$ the power.
Trap: Only independent sources are inputs. A dependent source is not.

WORK

Steps — scale a known answer

1. FACTOR
$K = \dfrac{\boxed{\text{new source}}}{\boxed{\text{old source}}}$

↓

2. SCALE
v, i: $K \cdot \boxed{\text{known}}$ $\quad$ p: $K^2 \cdot \boxed{\text{known}}$

Steps — unit output (ladders)

1. ASSUME
The output is 1 V (or 1 A).

↓

2. WALK BACK
Toward the source: $i = \frac{v}{R}$, KVL for the next node, KCL for the next current.

↓

3. SOURCE YOU FOUND
Call it v_s'.

↓

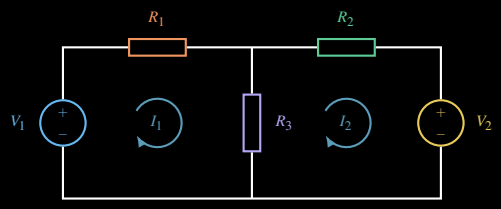
4. SCALE
$K = \dfrac{v_s}{v_s'}$. $\quad$ real output $= K \cdot 1$. Every value $\times K$.

17 — mesh-current analysis [book 3-3, 3-4]

Mesh currents and KVL

EXPLANATION

THE IDEA
Give each mesh one clockwise current. Write one KVL per mesh. A resistor between two meshes carries the difference of their currents.



Pieces

Piece	Where to find it
own R	resistors only in this mesh (R_1 for mesh 1)
shared R	a resistor on the border of two meshes (R_3)
rise	a source clockwise enters at $-$: $+v_s$
drop	a source clockwise enters at $+$: $-v_s$

Formula	In words
$i_{\text{shared}} = I_{\text{this}} - I_{\text{other}}$	current in a shared R

Trap: The shared R is in both equations, with the other mesh current subtracted.
Trap: Mesh analysis needs a planar circuit (no crossing wires).
Trap: A negative mesh current flows counterclockwise.

WORK

1. MARK THE MESHES
One clockwise I_1, I_2, … per mesh.

↓

2. READ THE PIECES PER MESH
own R, shared R (and with whom), sources as rise or drop.

↓

3. ONE KVL PER MESH (BY INSPECTION)
$$\left(\Sigma\boxed{\text{own}} + \Sigma\boxed{\text{shared}}\right) I_k - \Sigma\boxed{\text{shared with } j}\, I_j = \Sigma\boxed{\text{rises}}$$
(a drop enters the right side as $-v_s$)

↓

4. CHECK THE MATRIX
Symmetric: the I_2 number in row 1 equals the I_1 number in row 2. Off-diagonal entries are negative.

↓

5. SOLVE
Cramer, substitution, or a calculator (rref).

↓

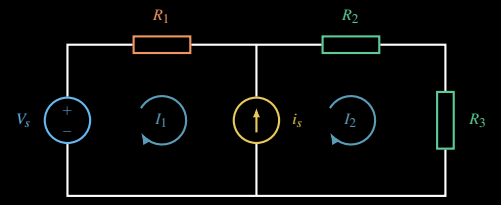
6. BRANCH CURRENTS
Own R: I_k. Shared R: $I_{\text{this}} - I_{\text{other}}$. Then $v = iR$.

↓

7. CHECK
KVL around one mesh with the numbers.

Current sources: known mesh current or supermesh

EXPLANATION



Pieces

Piece	Where to find it
i_s	the current source and its arrow
on the edge	the source is in only one mesh
shared	the source sits between two meshes (above)
outer loop	the path around both meshes, skipping the source branch

Dependent source	Do this
CCVS $r\,i_x$	write i_x in mesh currents ($I_1 - I_2$), put it in KVL, move it left
dependent current source	treat like i_s, then write its control in mesh currents

Trap: Never write KVL through a current source. Its voltage is unknown.

WORK

Steps — source on the outer edge

1. KNOWN CURRENT
$I_k = +\boxed{i_s}$ if the arrow is clockwise, $-\boxed{i_s}$ if not.

↓

2. SKIP ITS KVL
Write KVL only for the other meshes.

Steps — supermesh (shared source)

1. CONSTRAINT
$I_{\text{with the arrow}} - I_{\text{against}} = \boxed{i_s}$ *(above: $I_2 - I_1 = i_s$)*

↓

2. KVL AROUND THE OUTSIDE
Walk the outer loop of both meshes, skipping the source branch. Each R uses the mesh current it sits in.

↓

3. SOLVE
Substitute the constraint into the outer KVL.

↓

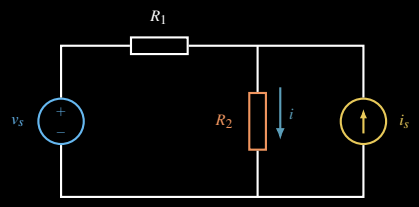
4. CHECK
The voltage across the source, found from the left and from the right, must match.

18–19 — source superposition, validating with simulation

18. Source superposition [book 3-5]

EXPLANATION

THE IDEA
In a linear circuit with several independent sources, any v or i is the sum of the answers from each source acting alone.



Pieces

Piece	Where to find it
the target	the v or i you want, with its reference arrow
voltage sources	each independent v_s
current sources	each independent i_s
dependent sources	stay on in every sub-circuit

Turn off	Replace with
voltage source	a short (wire), $v = 0$
current source	an open (gap), $i = 0$

Trap: Power does not superpose. Add the currents, then $p = i^2R$.
Trap: Keep one reference direction. A part flowing the other way enters the sum negative.

WORK

1. LIST THE SOURCES
Every independent v_s and i_s. Mark the target arrow.

↓

2. KEEP ONE, TURN OFF THE REST
v_s off → short. i_s off → open. Dependent: keep.

↓

3. SOLVE THE SMALL CIRCUIT
Find the part $\boxed{\text{target}}_k$. Often a divider (§10). Else nodal or mesh.

↓

4. REPEAT
Once per independent source.

↓

5. ADD
$\boxed{\text{target}} = \boxed{\text{part}}_1 + \boxed{\text{part}}_2 + \cdots$ *(all in the same direction)*

↓

6. POWER (IF ASKED)
From the total: $p = i^2R$. Never add the separate powers.

↓

7. CHECK
Solve the full circuit once with nodal or mesh. Same answer.

19. Validating with simulation [book 3-10]

EXPLANATION

THE IDEA
A simulator solves the same equations. Use it to check a hand answer, and to test that the circuit behaves linearly.

Formula	In words
$\%\,\text{error} = \dfrac{\lvert\text{sim} - \text{hand}\rvert}{\lvert\text{hand}\rvert} \times 100$	how far apart the answers are
DC operating point	every node voltage and branch current
sweep one source	linear: a straight line through the origin

Trap: A sign difference is usually a direction mismatch, not a wrong answer.
Trap: Round only at the end.

WORK

Steps — check a hand answer

1. SAME SETUP
Same ground node, same reference directions as the hand solution.

↓

2. RUN
DC operating point. Read the target v or i.

↓

3. COMPARE
% error. A negative reading means the current flows opposite to your arrow.

Steps — check linearity

1. SCALE TEST
Double one source. The output must double.

↓

2. SUPERPOSITION TEST
Run each source alone, then all on. The parts must add to the total.